Neural Tuning and Representational Geometry

Kriegeskorte & Wei (2021), Nature Reviews Neuroscience

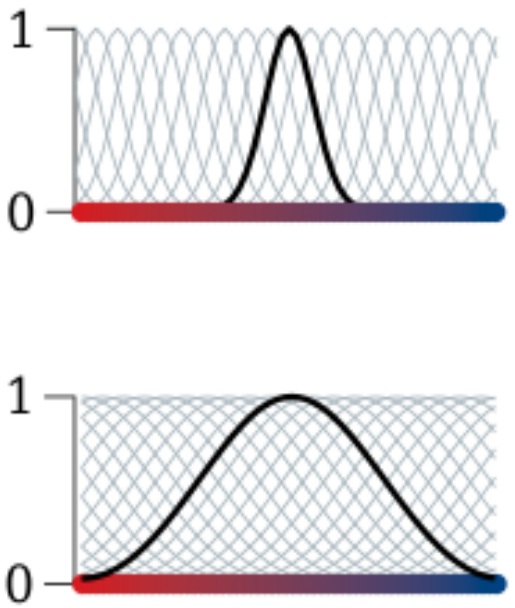
Presented by Audrey and Kevin

The Big Question

- When you see something, many brain cells fire. How does that activity stand for what you saw?
- Scientists usually answer this in one of two ways:
- View 1: look at one cell at a time and ask what it likes.
- View 2: look at the whole group of cells together and ask how the overall patterns compare.
- This paper explains how the two views are connected.

View 1: Each Cell Has a Favorite

- Many brain cells fire most for one "favorite" thing, such as a line tilted at one angle.
- Like a radio tuned to one station, the cell responds less as the line tilts away from its favorite.
- A plot of this is called a tuning curve (right).
- Top: a picky cell that only likes a narrow range of angles. Bottom: a less picky cell.



View 2: A Map of Patterns

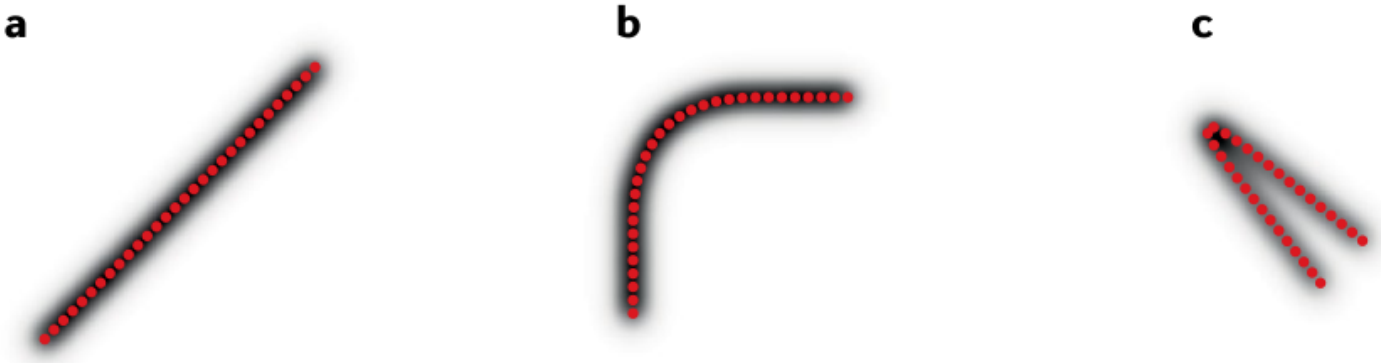
- Each picture you see produces a pattern of activity across all the cells.
- Compare every pair of patterns: similar patterns sit close together, different ones sit far apart.
- This gives a "map" of how the brain groups things. The paper calls it the representational geometry.
- The map tells us which things are easy for the brain to tell apart (far apart) and which are easy to mix up (close together).

Main Point 1: The Cells Build the Map, but Not the Other Way Around

- If you know what every cell likes, you can work out the map.
- But the same map can be built by very different sets of cells.
- Analogy: rotating a map changes which way is "up", but every distance stays the same.
- So the map tells you what information is there, but not which cells carry it.

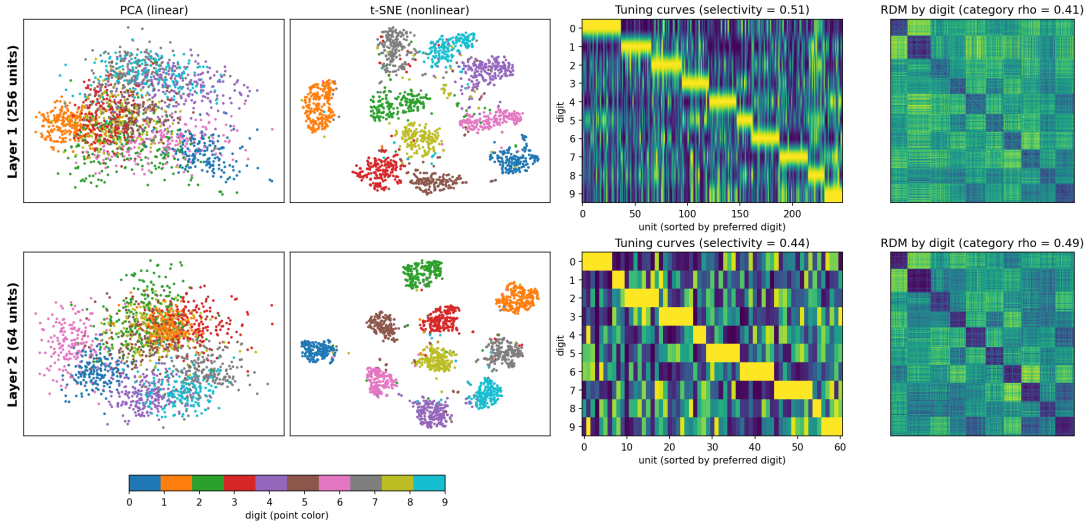
Main Point 2: Telling Things Apart

- Older work mostly measures how well the brain separates very similar things (e.g., two almost identical angles).
- Below: three maps (straight, bent, folded). Neighbouring points are equally far apart in all three.
- But when folded, opposite ends come close together, so very different things become easy to confuse.
- Lesson: you need the whole map, not just nearby comparisons.



Our Mini-Experiment: Inside a Digit-Recognizing Network

- We trained a small computer network to read handwritten digits (96% correct) and compared two of its layers.
- The deeper layer is simpler: it packs the information into far fewer directions (8 instead of 27).
- The deeper layer groups the same digits closer together on its map (match to "same digit" grouping rose from 0.41 to 0.49).
- But single units in the deeper layer are less picky about one digit, so the digit information is shared across the group.



Takeaways and Discussion

- Looking at single cells and looking at the whole group are two sides of the same coin.
- The group map tells you what the brain can tell apart; single cells tell you how it is built.
- Limitation: the paper is mostly theory with simple examples, not new experiments.
- Question for the group: when is it better to study one cell at a time, and when is the big picture enough?